
IntraChat
+main(String[] args)
boots GUI
Login
+JPanel Rootpanel
+JPanel jPanel3
+JTextField codeTF
+JPasswordField passwordF
+JTextField usernameTF
+JButton loginBtn
+JButton startSeverBtn
+JLabel username
+JLabel code
+JLabel password
+JLabel jLabel5
+JLabel jLabel2
+JLabel logo
+JLabel jLabel1
+codeTFActionPerformed(ActionEvent evt)
authenticates via socket

initializes connection

generates/verifies codes

connects to socket / authenticates

manages (0..*)

references back

queries & logs data

Server

-int port
-ServerSocket serverSocket
-boolean isRunning
+int defaultPort
-List<ClientHandler> clients
-Object stopLock
-boolean isStopped

+Server(int port)
+start()
+broadcast(String message, ClientHandler sender)
+sendMessageToUsername(String message, String username) bool
+broadcastUserList()
+removeClient(ClientHandler client)
+getClients() List<ClientHandler>
+getServerIpAddress() String
+getLocalLanIp()$ String
+getAllLanIps()$ List<String>
+getPort() int
+isRunning() boolean
+stop()
+getConnectionCode() String

Client

-String serverIp
-int serverPort
-Socket socket
-BufferedReader in
-PrintWriter out
-boolean isConnected
-Consumer<String> messageListener
-String lastErrorMessage
-Queue<String> pendingQueue

+Client(String serverIp, int serverPort)
+setMessageListener(Consumer<String> listener)
+String getLastErrorMessage()
+boolean connect(String username, String password)
-listenForMessages()
+sendMessage(String message)
+boolean isConnected()
+String getServerIp()
+int getServerPort()
+disconnect()

AuthService

-SecureRandom secureRandom
-long Day
-String HMAC_ALGORITHM
-String PBKDF2_ALGORITHM
-int PBKDF2_ITERATIONS

+byte[] generateSigningKey()
+String hashPassword(String password)
+boolean verifyPassword(String password, String stored)
+String generateCode(byte[] serverKey, String username, InetAddress s
+boolean verifyCode(byte[] serverKey, String code, String username, Ine
+String generateConnectionCode(String ip, int port)
+ConnectionTarget parseConnectionCode(String input)
+String[] splitCode(byte[] serverKey, String code)

ClientHandler

-Socket socket
-Server server
-BufferedReader in
-PrintWriter out
-String username
-String password

+ClientHandler(Socket socket, Server server)
+run()

«class»
DatabaseManager
-String DB_URL

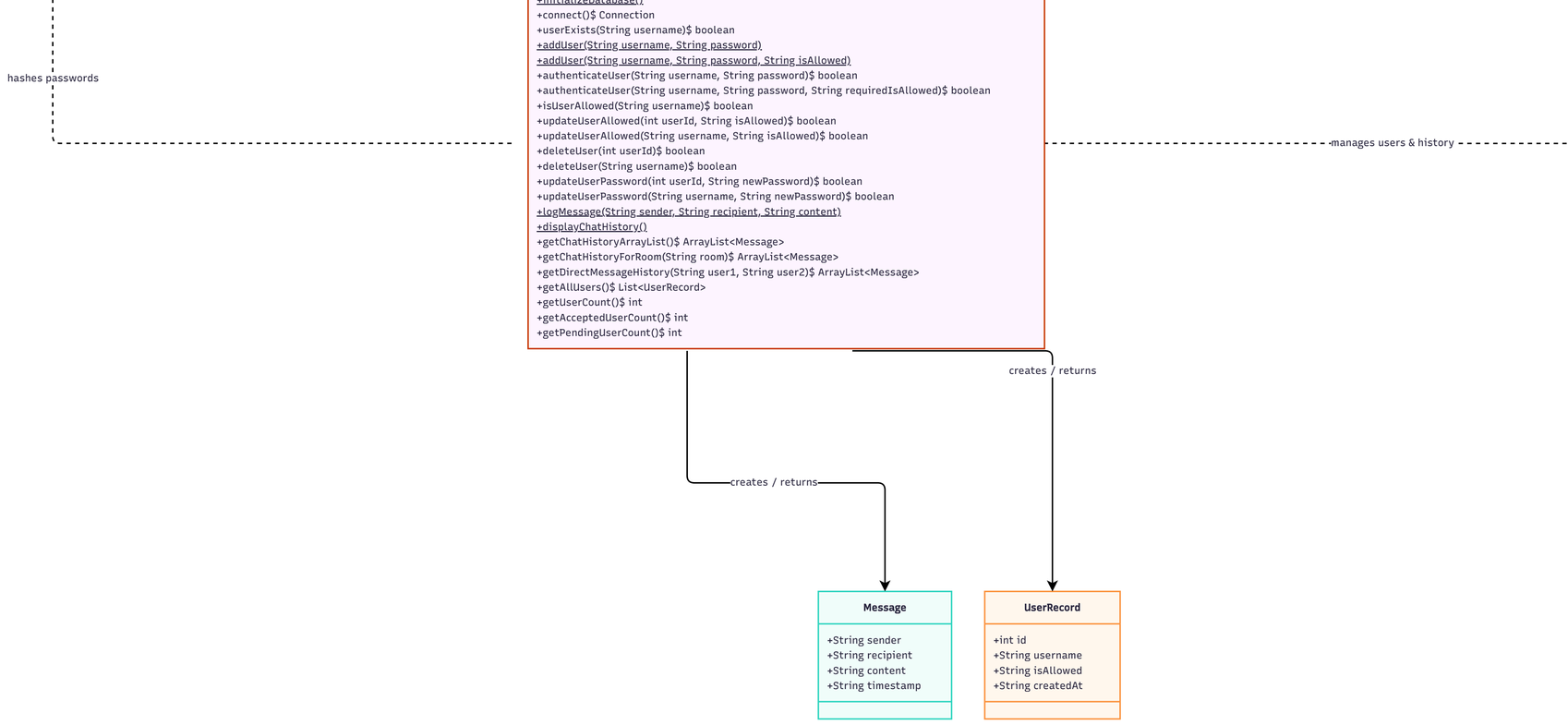
+initializeDatabase()

lean

controls state

AdminConsole

+JMenuBar jMenuBar1
+JPanel jPanel1
+JPanel jPanel2


hashes passwords
+initializeDatabase()
+connect()$ Connection
+userExists(String username)$ boolean
+addUser(String username, String password)
+addUser(String username, String password, String isAllowed)
+authenticateUser(String username, String password)$ boolean
+authenticateUser(String username, String password, String requiredIsAllowed)$ boolean
+isUserAllowed(String username)$ boolean
+updateUserAllowed(int userId, String isAllowed)$ boolean
+updateUserAllowed(String username, String isAllowed)$ boolean
+deleteUser(int userId)$ boolean
+deleteUser(String username)$ boolean
+updateUserPassword(int userId, String newPassword)$ boolean
+updateUserPassword(String username, String newPassword)$ boolean
+logMessage(String sender, String recipient, String content)
+displayChatHistory()
+getChatHistoryArrayList()$ ArrayList<Message>
+getChatHistoryForRoom(String room)$ ArrayList<Message>
+getDirectMessageHistory(String user1, String user2)$ ArrayList<Message>
+getAllUsers()$ List<UserRecord>
+getUserCount()$ int
+getAcceptedUserCount()$ int
+getPendingUserCount()$ int
manages users & history
creates / returns
creates / returns
Message
+String sender
+String recipient
+String content
+String timestamp
UserRecord
+int id
+String username
+String isAllowed
+String createdAt

+JScrollPane jScrollPane1
+JTable jTable1
+JTextField jTextField1
+JButton jButton1
+JButton jButton2
+JButton jButton3
+JButton jButton4
+JButton jButton5
+JButton jButton6
+JButton jButton7
+JButton jButton8
+JButton jButton9

+initComponents()
+jButton3ActionPerformed(ActionEvent evt)
+jButton9ActionPerformed(ActionEvent evt)